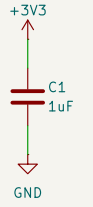
# CoffeeFlow — combined board, AC side (pump dimmer + valve SSR)

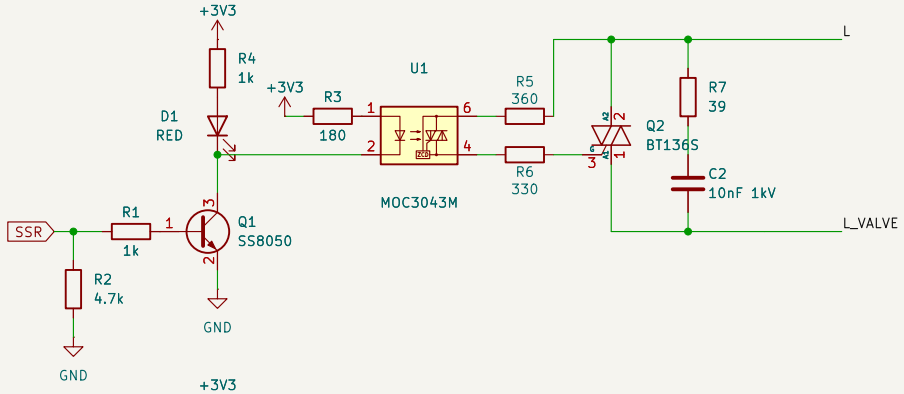
LOW VOLTAGE SIDE (3.3 V logic) | everything right of the optocouplers is MAINS 230 V — keep isolation slot / creepage under U1, U2, U3.

## 1 / Logic inputs

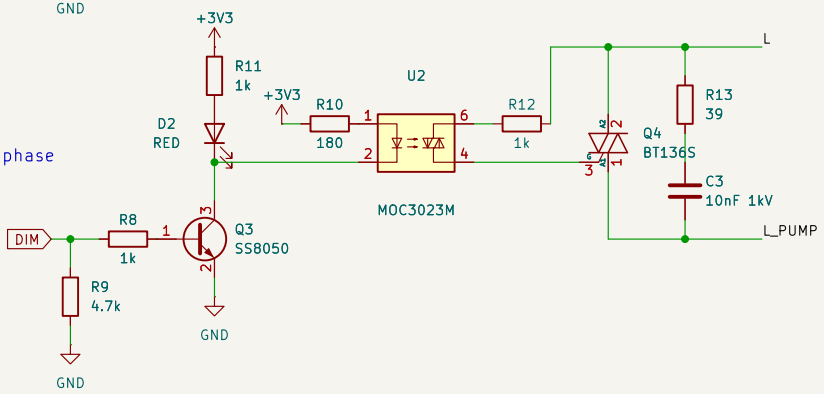
DIM, ZC, SSR and 3V3 from the XIAO (Controller sheet)



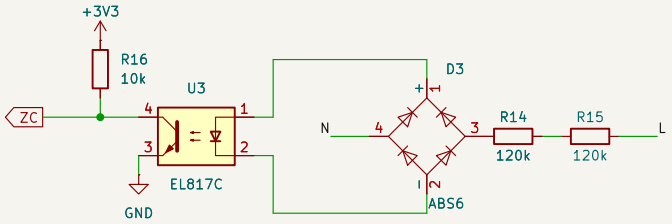

## 2 / Valve SSR — zero-cross (M5Stack Unit SSR, 3.3 V drive)




## 3 / Pump dimmer — random phase (RobotDyn 4A, ZC/DIM)

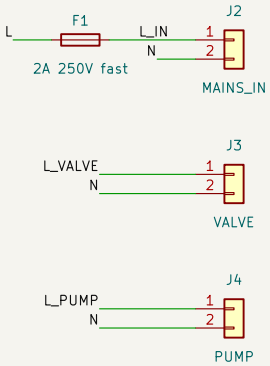



## 4 / Zero-cross detector




ZC is high around each zero crossing (opto off), low elsewhere.
With 240k and CTR >= 200 %: pulse about +/−0.5 ms, rising edge before the zero.

## 5 / Mains connectors (HT3.96)




N is common to all three connectors.
F1 protects both channels and the ZC sense.

Controller

File: combined_controller.kicad_sch

Interfaces

File: combined_interfaces.kicad_sch

**CoffeeFlow**

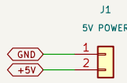
Sheet: /

File: combined.kicad_sch

***Title: Combined — AC side: pump dimmer, valve SSR, zero-cross***

| Size: A4 | Date: | **Rev:** |
|---|---|---|
| KiCad E.D.A. 10.0.6 | | Id: 1/3 |

## 1. POWER — external regulated 5V / XIAO onboard 3.3V regulator

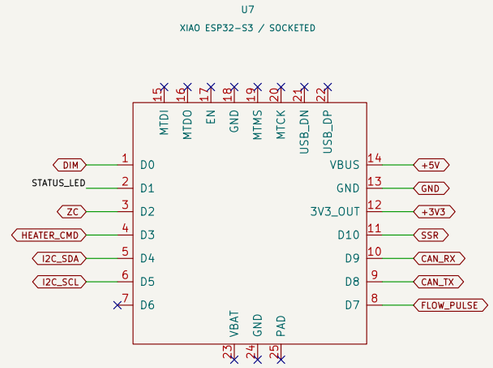



J1 feeds XIAO 5V and the 5V interfaces.
XIAO 3V3 supplies CAN VIO, I2C, TMP102, ADS1115, FLOW pull-up and the AC-side logic (SSR/DIM drivers, ZC pull-up).
USB-C, LDO, BOOT and RESET are on the removable XIAO.
Disconnect external J1 power before USB flashing.



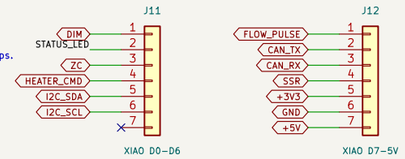

3V3 PWR_FLAG represents the regulator output on the XIAO.

## 2. XIAO ESP32-S3 — two female 1x7 sockets, 2.54mm pitch




U7: D0/GPIO1 DIM; D1/GPIO2 LED; D2/GPIO3 ZC; D3/GPIO4 BOILER.
D4/GPIO5 SDA; D5/GPIO6 SCL; D7/GPIO44 FLOW; D8/GPIO7 CAN TX.
D9/GPIO8 CAN RX; D10/GPIO9 SSR (valve).

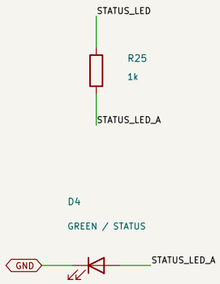
Seeed numbering: 1..7 = D0..D6; 8..11 = D7..D10.
12=3V3 OUT; 13=GND; 14=5V IN. Pins 15..25 are bottom pads, NC.
D6/GPIO43 unused.




Footprint: Combined_Local:XIAO-ESP32-S3-DIP-Headers.
Fit two female 1x7 sockets; XIAO fitted manually from stock.
Row spacing 15.24mm; 2.54mm pitch. GPIO4 avoids boot straps.

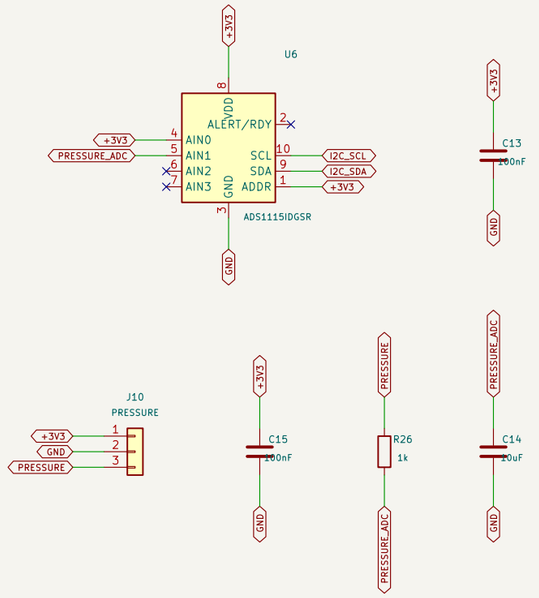
J11/J12: female 1x7 sockets soldered by JLCPCB (THT); the XIAO (U7) plugs in by hand and is not on the board. J11 pin n = XIAO pin n; J12 pin n = XIAO pin n+7.

## 3. STATUS LED — GPIO2 / D1, HIGH = ON




Existing green LED and 1k resistor retained.

## 10. PRESSURE ADC — optional ADS1115, J10 = screen_sensors J3 order




U6 ADS1115 on the shared I2C bus, ADDR tied to 3V3: 0x49 (TMP102 already uses 0x48).
A0 reads the local 3V3; A1 reads pressure through R26/C14 (1k / 10uF, 16 Hz, 10 ms).
A2, A3 and ALERT/RDY unused. J10 pin order matches screen_sensors J3: same cable.
XDB401 output 0.4–2.4V on 3V3 supply, not ratiometric: use the absolute A1 voltage.




H1: M2.5 mounting hole, 2.7mm NPTH, not connected.

FUTURE PCB FIRMWARE: BOILER GPIO4, active HIGH (MOSFET driver).
Preload LOW before output enable; disable heater pull-up.
Current HW-399 firmware uses GPIO3, active LOW: do not use unchanged.
Other existing XIAO pins retained; add status LED on GPIO2.
Firmware unchanged; PCB placement and routing pending.

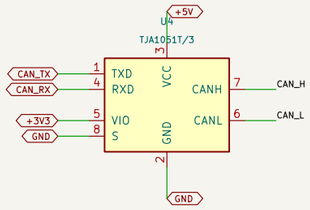
Socketed XIAO ESP32-S3 / Seeed DIP footprint
**CoffeeFlow**

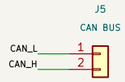
Sheet: /Controller/
File: combined_controller.kicad_sch

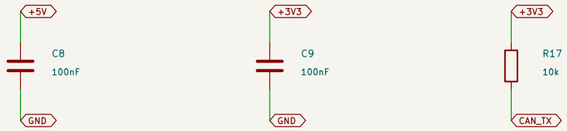
***Title: Combined — XIAO ESP32-S3, 5V input, status LED and pressure ADC***

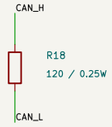
| Size: A3 | Date: 2026-10-13 | **Rev: A** |
|---|---|---|
| KiCad E.D.A. 10.0.6 | | Id: 2/3 |

## 4. CAN — TJA1051T/3, 5V bus driver / 3.3V logic



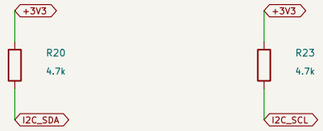



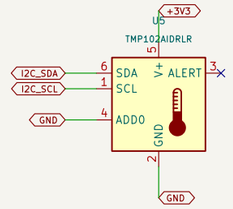




S/SLNT hard-wired low. R17 keeps TX recessive during reset.
R18 permanently terminates this bus-end node (120 ohms).
J5 has no GND pin: power GND must run alongside CAN pair.

## 5. I2C — 3.3V, on-board TMP102 and ADS1115



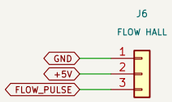







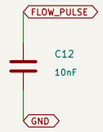
R20/R23 (4.7k) pull SDA/SCL up to 3V3: no off-board I2C connector.
Addresses: TMP102 0x48 (ADD0 grounded, ALERT unused) / ADS1115 0x49.
Mount the TMP102 away from XIAO and MOSFET heat.
100kHz initial bus speed.

## 6. DIGMESA — open collector, existing RC filter



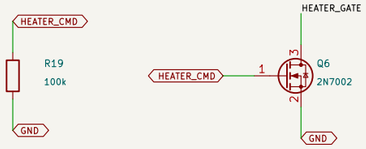


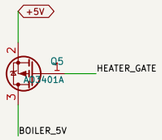


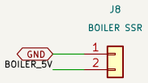



GPIO44 / D7 input; internal pull-up disabled. 1k x 10nF = 10us.
Digmesa powered from 5V; signal pulled to 3.3V only.
Pin order: GND brown / 5V green / SIGNAL yellow.
RC discharge speed also depends on sensor/cable impedance.

## 8. BOILER SSR — powered 5V output, HIGH command = ON










J8: SSR (-) -> pin1 GND; SSR (+) -> pin2 switched 5V.
Q6 pulls Q5 gate LOW; Q5 supplies the SSR current.
GPIO4 / D3 -> HEATER_CMD (HIGH = ON).
R19/R21/R22 ensure off with GPIO floating or 3V3 absent. Shared GND.
Gate pull-up current about 0.5mA when ON; load <=100mA; no mains.

**CoffeeFlow**

Sheet: /Interfaces/
File: combined_interfaces.kicad_sch

***Title: Combined — CAN, I2C, flowmeter and boiler SSR output***

| Size: A3 | Date: 2026-10-13 | **Rev: A** |
|---|---|---|
| KiCad E.D.A. 10.0.6 | | Id: 3/3 |